

Mobile Insight

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ABSTRACT

This project presents a comprehensive full-stack web application designed to streamline the process of gathering, analyzing, and presenting mobile reviews sourced from Reddit. The application aims to assist users in making informed decisions about vehicles by providing a detailed and categorized summary of user-generated reviews. By entering specific mobile details such as make, model, and year, users can retrieve an in-depth analysis of reviews, which are segmented into key aspects like performance, comfort, technology, reliability, and overall value.

The backend of the application is developed using Spring Boot, offering robust and scalable APIs for data handling and review aggregation. For data storage, MongoDB is utilized to manage both raw review data and processed summaries efficiently, leveraging its flexibility in handling unstructured and semi-structured data. On the frontend, React is employed to create an intuitive and interactive user interface, ensuring a seamless user experience.

The system leverages techniques such as text processing and sentiment analysis to categorize and synthesize reviews into meaningful insights. These insights are presented in an easy-to-understand format, enabling users to quickly evaluate the pros, cons, and overall sentiment toward a specific vehicle. This project demonstrates the effective integration of modern web technologies to solve real-world problems, providing a valuable tool for mobile buyers and enthusiasts to make data-driven decisions.

1. INTRODUCTION

In the modern era of technology, smartphones play a pivotal role in our daily lives, making it essential for users to make well-informed decisions when selecting a device. This project introduces a full-stack web application designed to streamline the process of gathering and synthesizing smartphone reviews. By leveraging Reddit's

r/Smartphones subreddit as a rich source of user-generated content, the application enables users to retrieve detailed insights and summaries about specific smartphone models.

Users can search for a smartphone by providing its brand, model, and year, such as "Apple," "13 Pro," and "2021." The application then fetches relevant discussions and reviews from Reddit and utilizes a powerful ClaudeAI system to process the data. Using a structured prompt, the AI generates a comprehensive summary of the fetched reviews, categorizing them into key aspects such as performance, design, battery life, camera quality, and overall user sentiment.

This summary is presented to the user in an intuitive and accessible format, allowing them to quickly understand the strengths, weaknesses, and overall reception of the smartphone. By combining cutting-edge technologies such as data aggregation, natural language processing, and AI-driven analysis, the application provides a robust tool for users to make data-driven smartphone purchasing decisions efficiently. Below is the detailed explanation of each key feature

1.1 REDDIT REVIEW CRAWLING

To enhance the functionality of the application, robust backend services were developed to fetch relevant mobile reviews directly from Reddit, based on user inputs such as brand, model, and year. These backend services were designed to efficiently interact with Reddit's API, retrieving user-generated content from the r/Smartphones subreddit and other relevant sources. The backend utilized advanced filtering techniques to ensure that the reviews and discussions retrieved were highly relevant to the search query. This process allowed the system to provide users with precise and comprehensive insights into the performance, features, and overall user satisfaction of specific smartphone models.

To ensure smooth operation and compliance with Reddit's API usage policies, rate-limiting mechanisms were implemented. This involved monitoring the frequency and volume of requests sent to the API, preventing excessive querying that could lead to service interruptions or violations of Reddit's terms of service. The rate-limiting logic ensured that the application adhered to these guidelines while maintaining optimal efficiency in data

retrieval. These measures not only upheld the reliability of the API integration but also contributed to a seamless and trustworthy user experience, enabling users to make well-informed decisions about mobile devices.

1.2 SUMMARIZATION

The application processes a vast number of mobile reviews gathered from multiple subreddits, including r/Smartphones, to generate structured and concise summaries that provide users with a clear understanding of a mobile device's overall reception. Leveraging advanced natural language processing techniques, the system synthesizes user-generated content into easily digestible insights. These summaries capture the essence of discussions and reviews, ensuring that users receive relevant and accurate information without the need to sift through extensive data manually.

The generated summaries are meticulously organized into meaningful sections such as performance, covering aspects like processing power, gaming capabilities, and multitasking; reliability, which includes durability, software updates, and long-term user satisfaction; and a pros and cons section that highlights the device's key strengths and potential drawbacks. This structured approach ensures that users can quickly evaluate the essential attributes of a mobile phone, aiding them in making informed decisions. By transforming unstructured data into actionable insights, the application serves as a reliable tool for anyone researching mobile devices.

1.3 SCALABILITY

The application was designed with scalability as a core consideration, ensuring that it can grow and adapt to future requirements with minimal disruption. A modular architecture was implemented, leveraging well-defined interfaces and design principles that promote separation of concerns. This architecture allows each component of the system—such as data fetching, processing, and summarization—to function independently, making it easier to extend or modify specific parts of the application without impacting the overall structure.

This modular approach is particularly advantageous for integrating additional data sources beyond Reddit, such as forums, tech blogs, or user review platforms. By adhering to interface-driven development, the application can seamlessly incorporate new sources with minimal changes to the existing codebase. Similarly, new features, such as enhanced sentiment analysis, device comparison tools, or interactive visualizations, can be integrated into the system with ease.

For mobile reviews specifically, this scalable design ensures that the application remains future-proof and capable of handling an increasing volume of data and user queries. Whether it's expanding the range of supported mobile brands and models or adding functionalities like regional review aggregation or feature-specific insights (e.g., camera performance or battery efficiency), the modular architecture enables the application to evolve alongside user needs and emerging technologies in the mobile industry.

1.4 ERROR MANAGEMENT

To ensure a robust and seamless user experience, a comprehensive error management system was implemented across both the backend and frontend of the application. In the backend, an exception handling framework was designed to address potential issues such as failed API requests, data processing errors, or connectivity problems. The framework categorizes errors into meaningful types, such as client-side errors (e.g., invalid input), server-side errors, and external API-related issues. Each error is accompanied by detailed, structured responses, ensuring the system provides consistent and meaningful feedback even during unexpected scenarios. This approach improves reliability and allows developers to easily diagnose and resolve issues.

On the frontend, intuitive error messaging was designed to provide users with clear and actionable feedback. For example, if a user enters invalid details, they are prompted to correct their input with specific suggestions. Similarly, if the application encounters issues fetching reviews—such as API rate limits being exceeded or network connectivity problems—users are informed of the situation with messages explaining the issue and suggesting next steps, such as retrying later. The error messages are styled consistently with the application's interface to maintain a professional and cohesive user experience. This dual-layered approach to error management ensures that users are never left in the dark, contributing to a trustworthy and user-friendly mobile review platform.

2. ARCHITECTURE

This section will elaborately discuss about the application front end and back-end architecture.

2.1 FRONT END

The frontend of the mobile review application is built using ReactJS, providing a dynamic and responsive user interface. The core feature is an intuitive input form where users can enter mobile details such as brand, model, and year (e.g., "Apple," "iPhone 13 Pro," "2021"). The form includes validation to ensure accurate and complete inputs, enhancing usability. This input data is then sent to the backend via an API call to fetch relevant reviews from Reddit. Fig 1 shows the forms:

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Mobile

Enter mobile brand (e.g., apple)

Mobile Model

Enter mobile name (e.g., 13pro)

Year

Enter year (e.g., 2025)

Get Car Reviews

Fig 1: user form

To improve user experience, the application features loading indicators to inform users while the data is being fetched. If the request is successful, the application dynamically displays the summarized reviews in well-organized sections such as performance, battery life, camera quality, and design. This categorization helps users quickly access and understand key insights. In the event of an error, user-friendly error messages are displayed, providing clear guidance, such as checking input details or retrying later.

The interface is designed with simplicity and efficiency in mind, ensuring that users can seamlessly navigate the application. Key elements like the input form, loading spinner, and summary display are styled for readability and accessibility, maintaining a professional yet user-friendly aesthetic. ReactJS’s capabilities ensure that the frontend is fast, responsive, and adaptable, delivering a smooth and engaging experience to users. Fig 2 below shows the front-end interface with summary of the desired mobile model.

2021 apple 13 pro - Review Summary

Based on the review texts available, I can provide a comprehensive analysis focusing on the Apple iPhone 13 Pro from 2021. Here's the detailed breakdown:

1. Overall Performance

• Exceptional speed and responsiveness with the A15 Bionic chip

• Smooth multitasking capabilities

• Strong gaming performance with 5-core GPU

• Notable improvements in general system responsiveness

2. Display and Design

• ProMotion 120Hz adaptive refresh rate display

• Excellent brightness (up to 1000 nits typical, 1200 nits HDR)

• High-quality build with stainless steel frame

• Smaller notch compared to previous models

• Available in Sierra Blue, Graphite, Gold, and Silver

3. Camera and Photography

• Significant improvements in low-light performance

• New Cinematic mode for video (1080p at 30fps)

• Enhanced macro photography capabilities

• Triple 12MP camera system with improved sensors

• Better optical image stabilization

4. Battery Life and Charging

• Exceptional battery longevity

Fig2: summary

2.2 BACK END

The backend of the mobile review application is developed using Java Spring Boot, providing a robust and scalable framework for handling user requests and processing data. At its core, the MobileReviewController manages incoming HTTP requests and routes them to the appropriate service layers. Key routes include /mobile-reviews/crawl, which initiates the process of fetching mobile reviews. The RedditCrawlerService is responsible for crawling Reddit for relevant mobile reviews using the JRAW (Java Reddit API Wrapper) library. This service constructs dynamic search queries based on the provided mobile brand, model, and year, retrieving up to 10 relevant posts per subreddit from sources such as r/Smartphones and brand-specific subreddits. It extracts review content from submission titles and self-text, filtering for meaningful data containing keywords like “review.”

To transform this raw data into actionable insights, the SummaryGenerationService processes the reviews and generates structured summaries tailored to the user’s query. By integrating with the Claude API, this service sends the aggregated review data alongside a detailed prompt, ensuring that the summaries are context-aware and categorized into meaningful sections like performance, design, and battery life. This modular backend design ensures seamless integration of future features, delivering accurate, high-quality insights to users. The following prompt is used to obtain the summary from Claude API:

String prompt = String.format("""
Provide a comprehensive analysis of %s %s %s reviews from multiple
Reddit sources.
Analyze and synthesize the following key aspects:

1. Overall Performance
- Speed and responsiveness
- Processor and RAM capabilities
- Gaming and multitasking performance

2. Display and Design
- Screen quality (brightness, resolution, and refresh rate)
- Build quality and durability
- Ergonomics and aesthetics

3. Camera and Photography
- Photo and video quality (daylight and low-light performance)
- Camera features (zoom, stabilization, and modes)
- Comparison with competitors' camera performance

4. Battery Life and Charging
- Battery longevity in typical use
- Charging speed and compatibility with fast charging
- Power efficiency during intensive tasks

5. Software and Features
- User interface and software experience
- Frequency and quality of software updates
- Unique features or standout functionalities

6. Connectivity and Network
- 5G and LTE performance
- Wi-Fi and Bluetooth connectivity
- Dual SIM or eSIM functionality

7. Value Proposition
- Price vs. features

- Comparison with similar models from competitors
- Longevity and resale value
8. Pros and Cons Summary
- Top 3 strengths
- Top 3 potential drawbacks
9. Overall Sentiment
- Aggregate user satisfaction
- Recommendation rating
Review Texts:
%s
""", year, mobileMake, mobileModel, truncatedReviewText);

The backend incorporates a `RateLimiterService` to efficiently manage API usage, preventing overloading or exceeding limits when interacting with external APIs like Reddit and Claude. It ensures compliance with usage policies and maintains system reliability by regulating the frequency of API calls. To handle errors effectively, the application employs a robust exception handling framework, including base exceptions for general issues and custom exceptions like `RateLimitExceededException` and `InvalidMobileDetailsException` for specific scenarios. A `GlobalExceptionHandler` ensures consistent error responses by mapping exceptions to appropriate HTTP status codes, providing clear feedback to users and enhancing the application's stability.

2.3 DATA MODEL

The data model for the mobile review application is built using a **MongoDB schema** to efficiently store and manage review data fetched from Reddit. The schema is designed to capture all essential information about each review in a structured format, ensuring smooth integration with backend services. The **MobileReview** schema includes fields for storing raw and summarized review content, along with metadata that maintains data integrity and supports efficient query operations.

Here's the detailed breakdown of the **Review** schema fields:

Field	Type	Description
<code>_id</code>	<code>ObjectId</code>	Unique identifier for each review document.
<code>title</code>	<code>String</code>	Indexed and unique title of the review fetched from Reddit.
<code>originalText</code>	<code>String</code>	Unprocessed raw review content retrieved from Reddit.
<code>summarizedText</code>	<code>String</code>	Structured summary of the review generated by the Claude API.
<code>lastUpdated</code>	<code>Date</code>	Timestamp of the last update to the document.

<code>contentChanged</code>	<code>Boolean</code>	Indicates if the review content has been modified since fetching.
<code>contentChanged</code>	<code>String</code>	Metadata for schema-to-model mapping in Spring Boot.

This schema is optimized to handle large volumes of mobile reviews while remaining adaptable to future enhancements. It ensures scalability, reliability, and the ability to deliver fresh, accurate summaries to users as new data is fetched and processed. The inclusion of metadata like `_class` allows for smooth integration with Spring Boot's data management features, supporting the application's modular and extensible architecture.

3. FUTURE ENHANCEMENTS

The mobile review application has several potential enhancements to further improve user experience and functionality. User Authentication is a key feature planned for future development, allowing users to create accounts, save their searches, and revisit previously generated summaries. This personalized experience would enable users to build a library of insights tailored to their interests and make informed decisions more effectively. Additionally, authentication could facilitate features like user preferences for specific brands or models, enabling quicker and more customized searches.

The application can also expand its capabilities through multi-source crawling, incorporating data from other platforms such as mobile review websites, forums, and tech blogs alongside Reddit. This would provide a more comprehensive view of user feedback and expert opinions. Advanced analytics is another enhancement that could enrich the user experience by offering detailed insights such as sentiment analysis, feature-based comparisons, and trend tracking across mobile models. To improve accessibility, the application could be integrated into a mobile app, enabling users to perform searches and review summaries on the go, ensuring a seamless experience across devices. These enhancements would significantly broaden the application's reach and value for users.

4. CONCLUSION

In conclusion, this project successfully demonstrates the integration of web scraping, data processing, and full-stack development to create a user-friendly platform for mobile reviews. By leveraging modern technologies such as React, Java Spring Boot, and Generative AI, the application provides a robust solution for summarizing mobile reviews based on user input. The seamless combination of data aggregation and AI-driven summarization ensures that users receive accurate, concise, and actionable insights about specific mobile models.

The application not only achieves its initial objectives of simplifying the review process but also lays a solid foundation for

future enhancements. Potential expansions include integrating additional review sources, incorporating advanced analytics like sentiment analysis, or leveraging other Generative AI models for deeper insights. This project serves as a powerful tool for users seeking reliable and data-driven information about mobile devices.

5. REFERENCES

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